

Gebran Kastoun

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EDUCATION

Cornell University

Ithaca, NY

BS in Electrical and Computer Engineering, Computer Science (GPA: 3.64/4.0)

May 2028

- **Relevant Coursework:** Power Electronics, Computer Architecture, Microelectronics, Circuits, Embedded Systems, Digital System Design with Microcontrollers, Digital Logic & Computer Organization

EXPERIENCE

SpaceX

May 2026 – Aug. 2026

Avionics Hardware Intern

Hawthorne, CA

- Designed two 4-layer PCB interface cards for flight avionics, implementing point-of-load regulation, differential signal interfaces, and DFM requirements to integrate replacement flight hardware at 80% lower cost
- Served as Responsible Engineer for a flight-critical reentry-vehicle avionics box, owning hardware development from requirements through integration and resolving electrical, mechanical, manufacturing, and software interface issues
- Bench-tested modified power-conversion hardware and updated two avionics units for compatibility with revised vehicle power requirements, performing electrical verification and supporting hardware integration
- Supported hardware failure investigation by reviewing PCB layout and isolating 6 candidate failure points for targeted bench testing
- Owned testing for an internally developed Xpedition Designer MCP Server to improve schematic design time by 25% and reduce peer review time by 40%; presented findings to 250 employees, garnering 40 new users in 2 weeks

Cornell CUAir – 2nd Place, 2025 C-UASC Competition

Nov. 2024 – Present

Electrical Subteam Lead

Ithaca, NY

- Designed an ATmega328P gimbal stabilization PCB with IMU, servo, UART, and debug interfaces, reducing system size and weight by 50% and delivering CUAir's first functional gimbal in 6 years
- Performed board bring-up and validation using oscilloscopes and multimeters, debugging power, I²C, UART, and PWM interfaces; incorporated LEDs, jumper pads, and test points to accelerate integration and troubleshooting
- Developed C++ firmware for closed-loop stabilization using I²C IMU feedback, PID control, and PWM servo actuation, while streaming live gimbal angle telemetry over UART through a Raspberry Pi to the ground station
- Own aircraft electrical and avionics architecture, leading system integration and design reviews across 9 projects while mentoring 12 members and coordinating electrical, mechanical, and software interfaces

Cornell University

Jan. 2026 – Present

Teaching Assistant

Ithaca, NY

- **ECE 4760 – Digital Systems Design Using Microcontrollers:** Support 80 students debugging RP2350 systems in C across peripherals, real-time firmware, and HW/SW integration in 8 weekly lab hours
- **ECE 2100 – Circuits for Electrical & Computer Engineers:** Supported 120 students in circuit analysis and debugging through 4-hour weekly labs, office hours, grading, and technical guidance

PROJECTS

12S → 24 V Avionics Power Converter | Altium, Power Electronics, Synchronous Buck, DroneCAN

- Designed, laid out, and validated a 144 W 12S-to-24 V synchronous buck converter achieving 97% efficiency at 6 A, eliminating CUAir's dedicated 6S avionics battery by powering flight electronics from the main flight battery
- Engineered transient protection and input filtering for motor-induced bus disturbances, integrating input/output current sensing and streaming telemetry over DroneCAN to the Pixhawk flight controller and the ground station

TinyRV2 Five-Stage Pipelined Processor | Verilog, Computer Architecture, Quartus Prime

- Implemented a five-stage TinyRV2 processor in Verilog for a 27-instruction RISC-V subset, with data forwarding, load-use stalls, and branch squashing to handle data and control hazards
- Implemented valid/ready instruction and data memory interfaces with backpressure handling; verified RTL against a functional ISA model using directed and randomized Python/pytest tests and cycle-level pipeline traces

SKILLS

Hardware & Tools: Altium, Xpedition, Valor NPI, LTspice, Quartus Prime, Oscilloscopes, Logic Analyzers, DMMs, Bench Power Supplies, Electronic Loads, Soldering

Hardware Design: Schematic Capture, PCB Layout, Board Bring-Up, Hardware Validation, Power Delivery, Power Electronics, FPGA/Digital Design, DFM, Systems Integration

Programming Languages & HDL: Verilog/SystemVerilog, C/C++, Rust, Java, Python, Assembly

Embedded Systems: RP2040/RP2350, ATmega328P, FRDM-KL46Z, Raspberry Pi, Pixhawk, DroneCAN